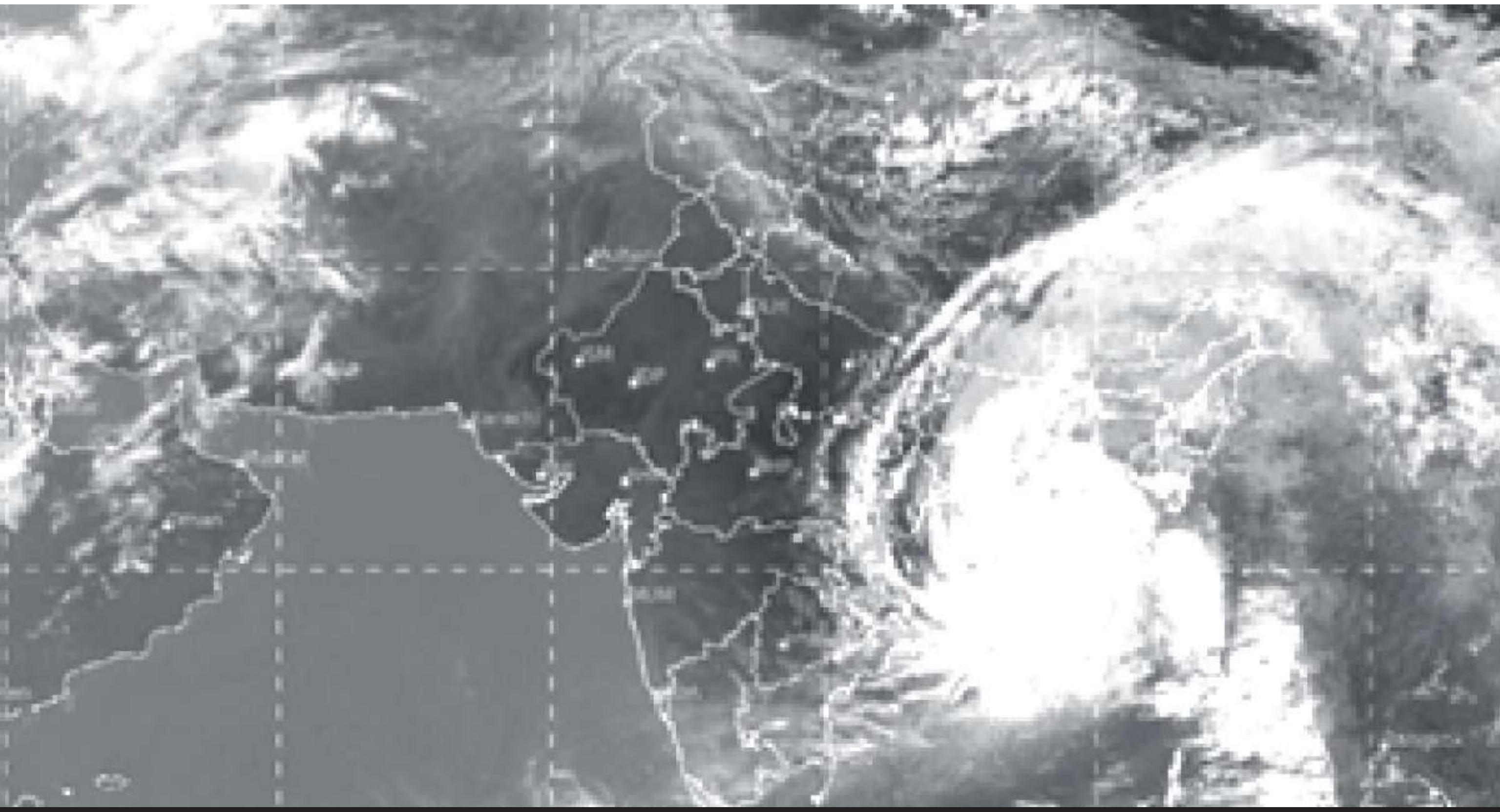
3 Coordinate Geometry



To accurately identify a location, a coordinate framework is required. When weather forecasters track hurricanes, they periodically note the exact location to see the storm's route and try to predict the storm's future path in the part based on these findings..

What's Inside

Topic(s)

- ✓ Topic-wise Notes
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- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
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- ✓ Detailed Explanation

TOPIC 1

INTRODUCTION TO COORDINATE GEOMETRY

Learning Objectives

- Students will be learning the main concepts of coordinate geometry.

Learning Outcomes

- Students will be able to identify the x and y coordinate of a point.

Real Life Application

We all have seen the aeroplanes flying in the sky but might have not thought of how they actually reach the correct destination. Actually all these air traffic is managed and regulated by using coordinate geometry. How much time is required for travelling from one place to another at certain speed? All the air traffic is controlled by air traffic controller.

A controller must know the location of every aircraft at any particular instant of time in the sky. Coordinates are used to describe current location of the aircraft. Even if an aircraft moves a small distance (up, down, forward or backward), its coordinates are updated in the system.

Now imagine what if coordinate system does not exist. Pilots, aircraft controllers, passengers in the flight and persons waiting for the flight will not be able to get the location or position of aircraft. This will also definitely increase the chances of aircraft crashes.

So from here we can easily say that coordinate system is one of the most important parts of air transport.

Let us now study and learn coordinate geometry in detail.

Coordinate geometry is a system of geometry where the positions of point on a plane are described by ordered pairs of numbers.

It deal with the position of an object lying in a plane described with the help of two mutually perpendicular lines.

Example 1. **A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South direction and East-West direction.**

All the other streets of the city run parallel to these roads and are 200 m apart. There are 5 streets in each direction. Using 1 cm = 100 m, draw a model of the city on your notebook to represent the roads/streets by single lines.

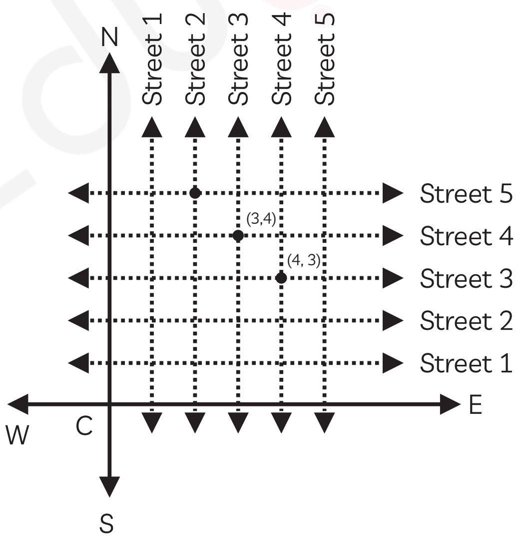
There are many cross-streets in your model. A particular cross-street is made by two streets, one running in the North-South direction and another in the East-West direction. Each cross-street is referred to in the following manner: If the 2 street running in the North-South direction and 5 in the East-West direction meet at some crossing, then we will call this cross-street (2, 5). Using this convention, find:

(A) How many cross-streets can be referred to as (4, 3)?

(B) How many cross-streets can be referred to as (3, 4)? [NCERT]

Ans. Model of street plan of the city:

Taking: 1 cm = 100 m



From the above diagram,

(A) Only one cross-street (Street 3 running North-South direction and street 4 running East-West direction) is referred to (3, 4).

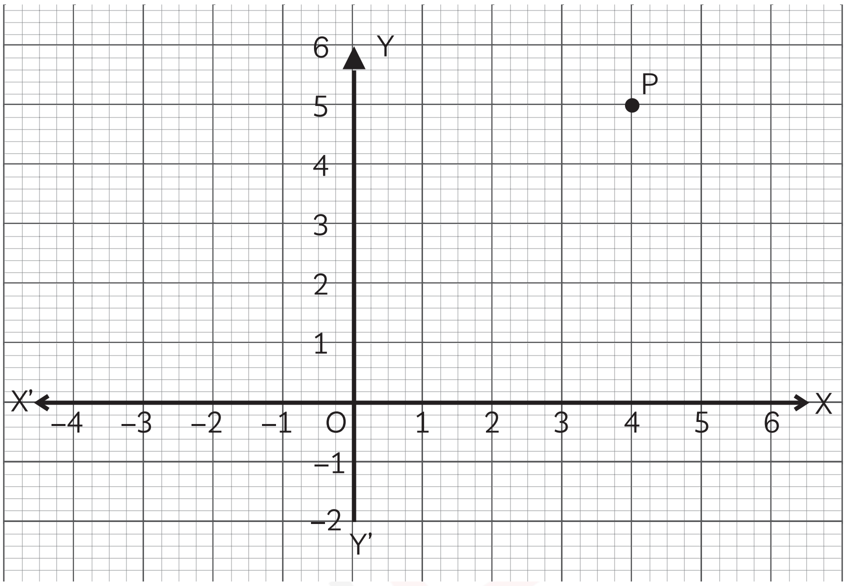
(B) Only one cross-street (Street 4 running North-South direction and street 3 running East-West direction) is referred to (3, 4).

OBJECTIVE Type Questions

Multiple Choice Questions

[1 mark each]

1. The coordinates of the point P shown in the given graph, is:



(a) (4, 5) **(b) (–4, 5)**

(c) (5, 4) **(d) (5, –4)**

Ans. *(a) (4, 5)*

Explanation: The point *P* is at a distance of 4 units to the right of *y*-axis. So, its abscissa will be 4. Also, the point P is at a distance of 5 units above the *x*-axis. So, its ordinate will be 5. Thus, the coordinates of the point *P* are (4, 5).

2. The coordinates of a point whose ordinate is 4 and lies on *y*-axis is:

(a) (4, 0) **(b) (0, 4)**

(c) (1, 4) **(d) (4, 2)** **[NCERT Exemplar]**

Ans. *(b) (0,4)*

Explanation: Any point which lies on the *y*–axis has the coordinates (0, *y*) when *y* can take any value. Here, *y* = + 4 units.

∴ The coordinates of the point are (0, 4).

Fill in the Blanks

[1 mark each]

3. **The point at which the two coordinate axes meet is called ....** **[NCERT Exemplar]**

Ans. *origin*

Explanation: Origin is the point where x-axis and y-axis intersect each other perpendicularly.

4. **Point (3, 0) lie in the first quadrant.** **[NCERT Exemplar]**

Ans. *False*

Explanation: The sign of abscissa and ordinate that lie in the first quadrant is both positive, but the point (3, 0) has ordinate as zero and that means the point lies on the x-axis and not in any quadrant, so, the answer of the given statement is false.

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Assertion-Reason

[A-R]

[1 mark each]

In the following question, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c) Assertion (A) is true but reason (R) is false.

(d) Assertion (A) is false but reason (R) is true.

5. **Assertion (A): Point P(1, –2) lies in IV quadrant.**

   Reason (R): In the Cartesian system, x and y coordinates of IV quadrant are positive and negative respectively.

Ans. *(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).*

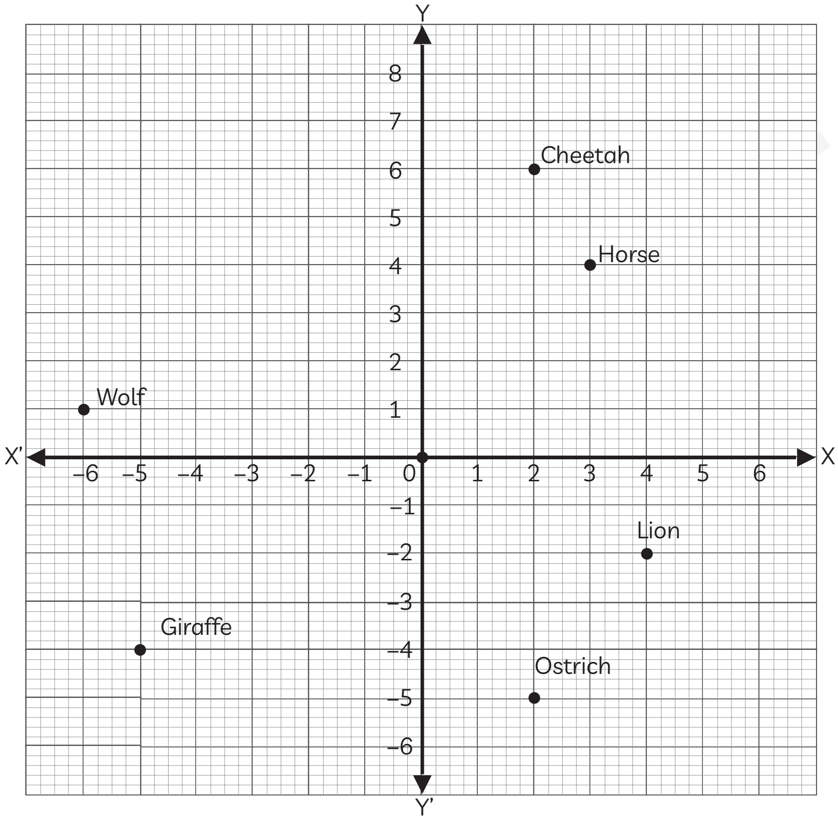
Explanation: The coordinates of point which lies in IV quadrant has positive x-coordinate and negative y-coordinate.

CASE BASED Questions **(CBQs)**

[**5** marks each]

Read the following passages and answer the questions that follow:

6. *In Jan 2020, Australia witnessed the worst-ever wildfire. Accordingly to various reports, about 50 crores animals and birds died or were seriously damaged. Thousands of people left their homes and the property of millions of dollars was damaged. If the forest is divided into four quadrants and animals have been placed at different places as shown in the figure.*



(A) Which among the following animals are collinear on the graph?

(a) Horse and Lion **(b) Giraffe and Ostrich**

(c) Lion and Giraffe **(d) Cheetah and Ostrich**

(B) If Ostrich travels 11 units towards the North direction, he will meet:

(a) Giraffe **(b) Horse**

(c) Cheetah **(d) Wolf**

(C) **If Giraffe moves 6 units towards East direction, he will be in:**

(a) **I quadrant** (b) **II quadrant**

(c) **III quadrant** (d) **IV quadrant**

(D) **Find the Abscissa of coordinates of the wolf.**

(a) **–6** (b) **1**

(c) **–7** (d) **–5**

Ans. (A) *(d) Cheetah and Ostrich*

Explanation: Collinear points are the points that lie on the same straight line, so we can analyze from the graph that cheetah and ostrich are collinear.

(B) *(b) Cheetah*

Explanation: Ostrich moves 11 units in North direction , i.e. along y-axis. So, ostrich will meet cheetah and the new coordinates of ostrich are (2,6).

(C) *(d) IV quadrant*

Explanation: The coordinates of giraffe are (–5, –4), Now moving 6 units towards East direction along x–axis so, Giraffes new coordinates are (–5 + 6, –4) = (1,–4)

Hence, Giraffe will be in IV quadrant.

(D) *(a) –6*

Explanation: The coordinates of the wolf are (–6, 1)

So, Abscissa = –6

VERY SHORT ANSWER Type Questions **(VSA)**

[**1** mark each]

7. In which quadrants do the points (–5, 2) and (2, –5) lie?

[NCERT Exemplar]

Ans. Signs of the coordinates of a point in the first quadrant are (+, +).

In the second quadrant (–, +).

In the third quadrant (–, –)and

In the fourth quadrant (+, –)

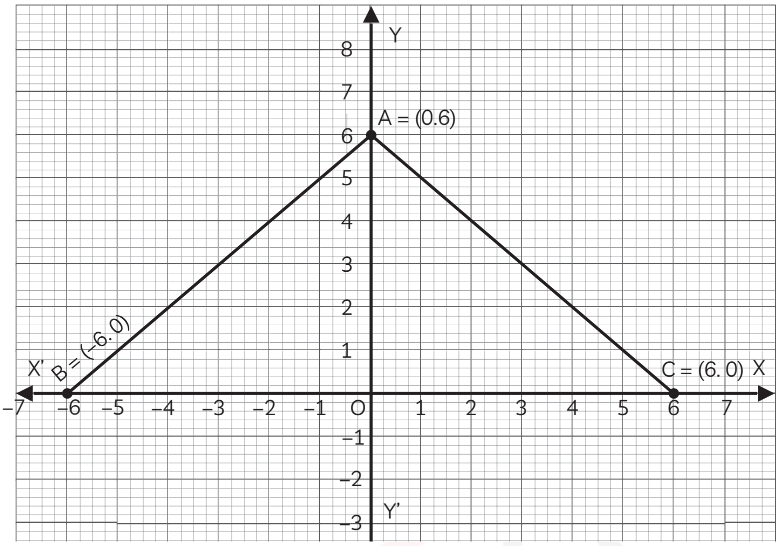
So, (–5, 2) lies in the second quadrant (–, +) and

(2, –5) lies in the fourth quadrant (+, –).

SHORT ANSWER Type-I Questions (SA-I)

[**2** marks each]

8. In the given graph, find the area of triangle ABC.



Ans. As per the given graph, points are A(0, 6),B(–6, 0) and (6, 0)

$$\text{Area of triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times BC \times OA$$

As, $BC = 6 - (-6)$

$= 12$ units

$OA = 6$ units.

$$\therefore \quad \text{Area of triangle} = \frac{1}{2} \times 12 \times 6$$

$= 36$ sq. units.

SHORT ANSWER Type-II Questions (SA-II)

[**3** marks each]

9. Find the coordinates of the vertices of a rectangle whose length and breadth are 8 and 4 units respectively, one vertex is at the origin, the longer side lies on the *y*–axis and one of the vertices lies in the second quadrant.

Ans. Let the rectangle be OABC with the vertex O at the origin.

Let OA be the longer side.

$\therefore$ The coordinates of O are (0, 0). Since, one of the vertices is in the second quadrant and the longer side of length 8 units lie on the y–axis, the coordinates of A is (0, 8).

Also, the rectangle lies in the second quadrant.

$\therefore$ The side OC will lie on x-axis to the negative direction.

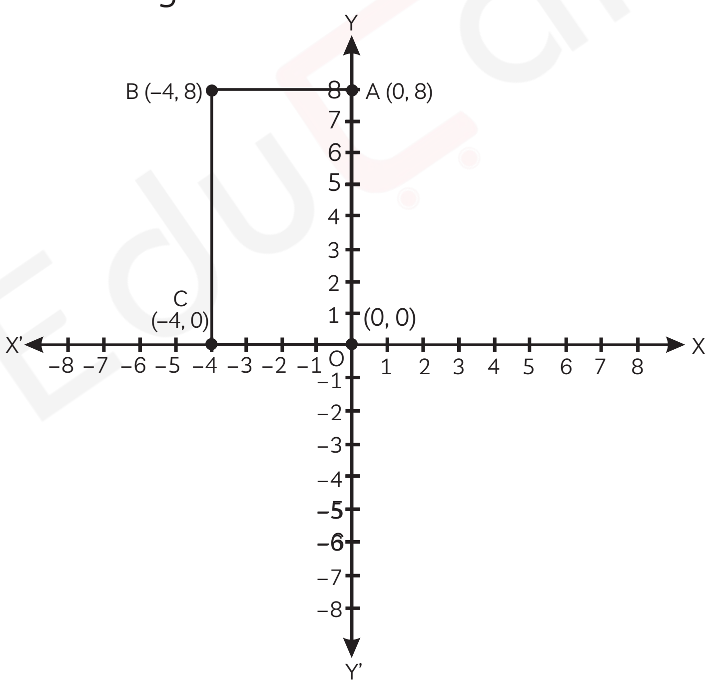
Then, the length of OC is 4 units.

$\therefore$ The coordinates of C will be (–4, 0)

Then obviously the coordinates of B will be (–4, 8).

Here, OC || AB and $\angle AOC = 90°$.

Hence, it is a rectangle.

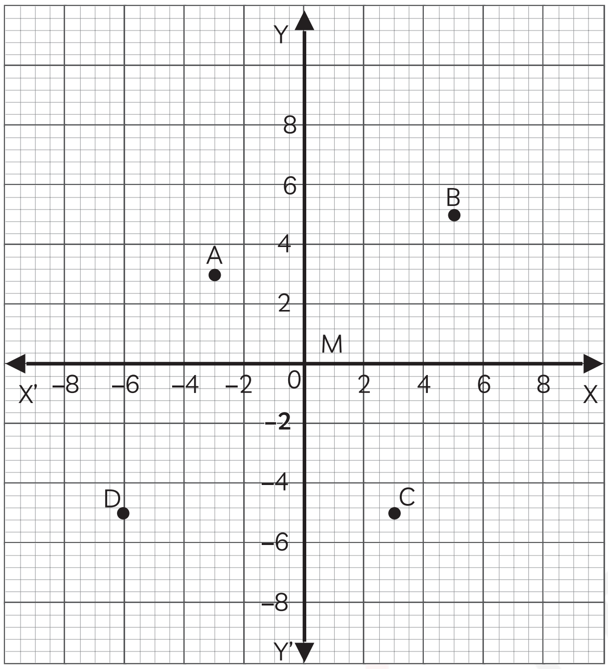


Therefore, the vertices of the rectangle are (–4, 8), (0, 8), (0, 0), and (–4, 0).

LONG ANSWER Type Questions (LA)

[**4** & **5** marks each]

10. Shranya draw a graph (as shown in the figure) and asked her friend to find:



(A) Coordinate with *x* and *y*-axis equal.

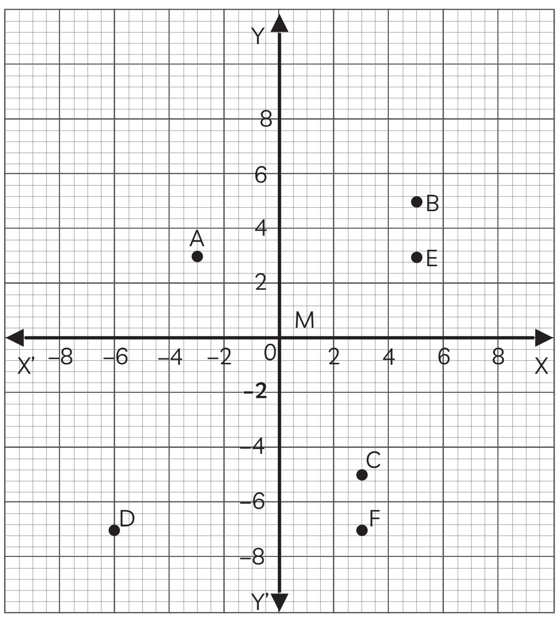
(B) Name the figure formed by joining ABCD.

(C) Coordinates having ordinate lesser than its abscissa.

Ans. (A) By looking at the graph we can see that point B has coordinates of *x* and *y* equal. *i.e.*, B(5, 5).

(B) By looking at the graph, the figure formed by the coordinates A, B, C and D can be found by finding the distance between coordinates A(–3, 3), B(5, 5), C(3, –5) and D(–6, –5).

To find distance of line AB, name coordinate E (5, 3)



Now, In ΔABE, apply Pythagoras theorem,

$$AB^2 = BE^2 + AE^2$$
$$= 2^2 + 8^2$$
$$= 4 + 64$$
$$AB^2 = 68$$
$$AB = 2\sqrt{17} \text{ units}$$

To find the distance of line CD, name coordinate F (3, –7)

Now, In Δ CDF, apply Pythagoras theorem,

$$CD^2 = CF^2 + DF^2$$
$$= 2^2 + 9^2$$
$$= 4 + 81 = 85$$
$$CD = \sqrt{85} \text{ units}$$

As, $AB \neq CD$

So, ABCD is neither a rectangle nor a parallelogram.

$\therefore$ ABCD is a quadrilateral.

(C) A(–3, 3), B(5, 5), C(3, –5) and D(–6, –7)

Only points C and D have ordinate lesser than abscissa *i.e.,*

$$C(3, -5) \rightarrow -5 < 3$$
$$D(-6, -7) \rightarrow -7 < -6$$